



SET BY: JH

MV
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LT

KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2004
TRIAL HSC EXAMINATION

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this paper
- All necessary working should be shown in every question

Total marks (84)

- Attempt Questions 1–7
- All questions are of equal value
- Use a **SEPARATE** Writing Booklet for each question
- Please write your **Board of Studies** Student Number and Teachers Initials on the front cover of each of your writing booklets.

NAME: _____

TEACHER: _____

Total marks (84)

Attempt questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)	Use a SEPARATE writing booklet	Marks
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(a)	Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{4x}$.	2
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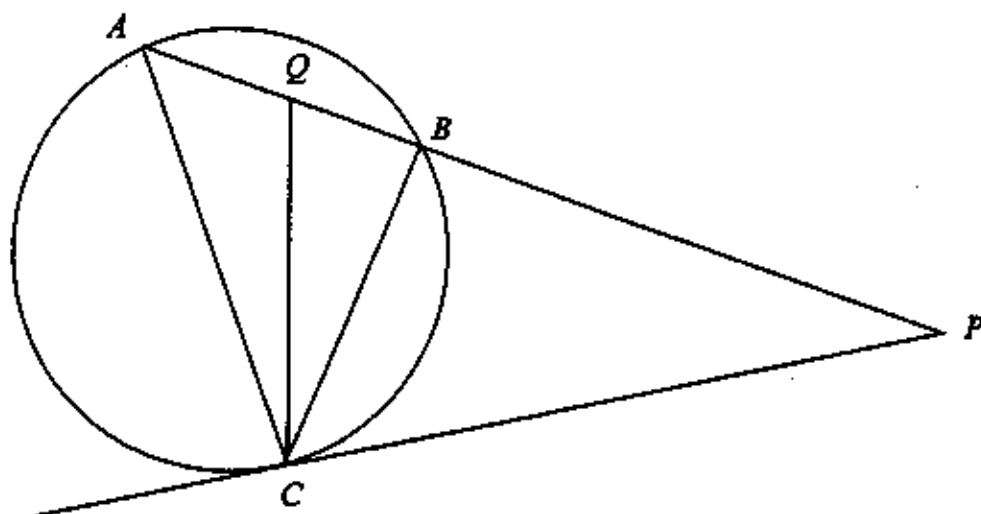
(b)	Find the exact value of $\int_2^3 \left(\frac{x^2}{x^3 - 7} \right) dx$.	3
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(c)	Solve for x : $\frac{2x}{x-1} \leq 1$.	3
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(d)	Find $\frac{d}{dx} \left(\tan^{-1} \frac{x}{3} \right)$.	1
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(e)	The point $P(19, -15)$ divides an interval AB externally in the ratio 3:2. Find the coordinates of the point $B(x, y)$ given $A(-2, 3)$.	3
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(a)



In the diagram above, PC is a tangent to the circle at C and QC bisects $\angle ACB$.

3

Copy the diagram into your writing booklet.

Prove, with reasons, that $PC = PQ$.

(b) Use the substitution $u = e^x$ to find: $\int \frac{dx}{e^x + 4e^{-x}}$

3

(c) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 2x \, dx$.

3

(d) Find the exact value of $\cos^{-1}\left(\sin \frac{4\pi}{3}\right)$.

3

Question 3 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) Find the value of the term independent of x in the expansion of $\left(x - \frac{2}{x^3}\right)^{12}$. 2
- (b) (i) Find the equation of the tangent to the curve $y = x^2 - x$ at the point where $x = 2$. 2
- (ii) Find the obtuse angle between the line $\frac{x}{3} + \frac{y}{2} = 1$ and the tangent found in part (i). Give your answer to the nearest degree. 2
- (c) (i) Express $\sqrt{12} \sin x + 2 \cos x$ in the form $A \cos(x - \alpha)$; where $A > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Hence, sketch the graph of $y = \sqrt{12} \sin x + 2 \cos x$, in the domain $0 \leq x \leq 2\pi$. 2
- (iii) State the number of solutions that satisfy the equation $\sqrt{12} \sin x + 2 \cos x = 1$ in the domain $0 \leq x \leq 2\pi$. 1
- (iv) Write down the general solution to $\sqrt{12} \sin x + 2 \cos x = 1$ 1

Question 4 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) Use one application of Newton's method to find a better approximation to the root of the equation $e^{-x} - \log_e x = 0$, given that there is a root near $x = 1.4$. Give your answer to 3 decimal places. 3
- (b) Use the Principle of Mathematical Induction to show that the expression $7^n + 5$ is divisible by 6 for all positive integers n . 4
- (c) (i) Find $\frac{d}{dx} \left(x \sin^{-1} \frac{x}{4} + \sqrt{16 - x^2} \right)$. 3
- (ii) Hence, evaluate $\int_0^4 \sin^{-1} \frac{x}{4} \, dx$. 2

- (a) Newton's Law of Cooling states that when an object at temperature T ($^{\circ}\text{C}$) is placed in an environment at a temperature R ($^{\circ}\text{C}$), then the rate of temperature loss is given by the equation $\frac{dT}{dt} = k(T - R)$; where t is the time in seconds and k is a constant.

A packet of peas, initially at 24°C is placed in a snap-freeze refrigerator in which the internal temperature is maintained at -40°C . After 5 seconds the temperature of the packet is 19°C . Suppose $T = R + Ae^{kt}$, where A is a constant.

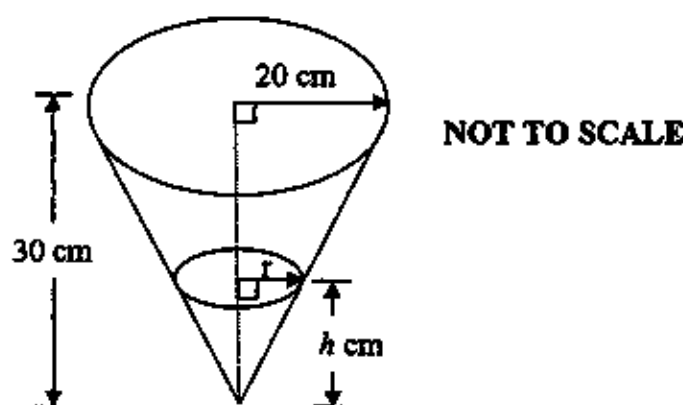
- (i) State the value of A . 1

- (ii) Show that $k = \frac{1}{5} \log_e \left(\frac{59}{64} \right)$. 2

- (iii) Hence show that it will take approximately 29 seconds for the packet's temperature to reduce to 0°C . 2

- (b) Prove that: $\tan \left(\frac{\pi}{4} + \theta \right) - \tan \left(\frac{\pi}{4} - \theta \right) = 2 \tan 2\theta$ 3

(c)



Water is poured into a conical vessel, of base radius 20 cm, and height 30 cm at a constant rate of 24 cm^3 per second. The depth of water is h cm at time t seconds and V is the volume of the water in the vessel at this time.

- (i) Explain why $r = \frac{2h}{3}$. 1

- (ii) Hence show that the volume of water in the vessel at any time t is given by $V = \frac{4\pi h^3}{27}$. 1

- (iii) Find the rate of increase of the area (A) of the surface of the water, when the depth is 16 cm. 2

(a) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$ ($a > 0$).

(i) By derivation, show that the equation of the chord is:

2

$$y = \frac{1}{2}(p+q)x - apq.$$

(ii) If the chord PQ passes through the focus, S , show that $pq = -1$.

2

(iii) Using the fact that $PQ = PS + SQ$, or otherwise, show that the chord PQ

3

has length $a\left(p + \frac{1}{p}\right)^2$.

(b) A particle moves along a straight line such that its distance from the origin at time t (s) is x (m) and its velocity is v .

(i) Prove that $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$.

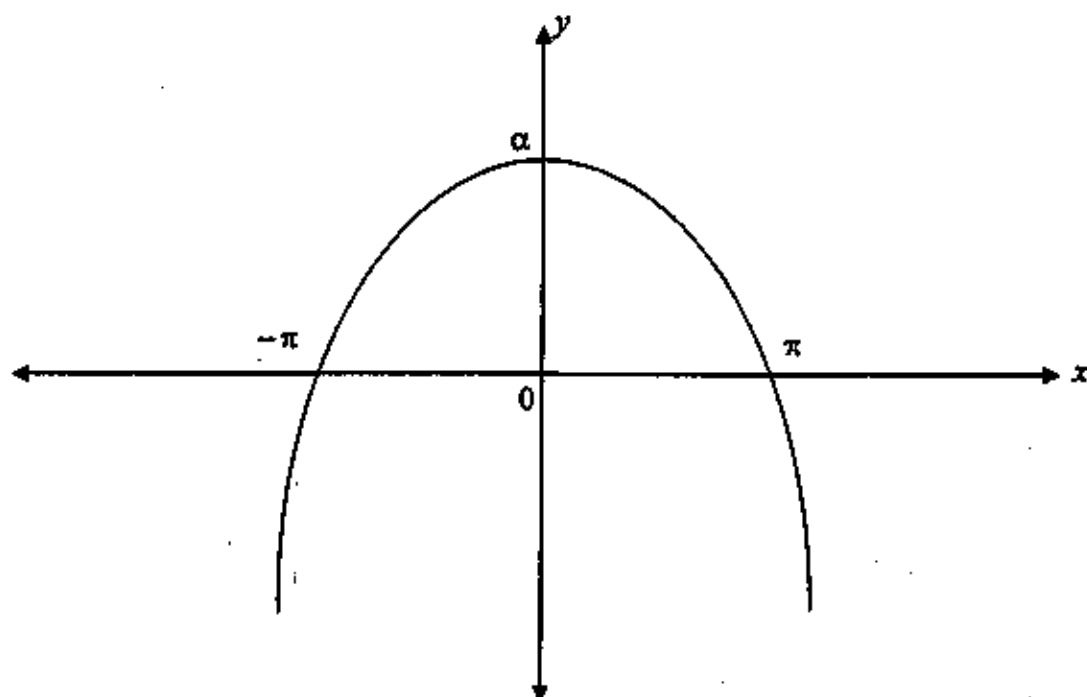
2

(ii) If the acceleration satisfies $\frac{d^2x}{dt^2} = -4\left(x + \frac{16}{x^3}\right)$ and the particle is

3

initially at rest when $x = 2$, show that $v^2 = 4\left(\frac{16 - x^4}{x^2}\right)$.

(a)



The diagram shows a parabola $y = f(x)$, with vertex $(0, \alpha)$ and $\alpha > 0$. The parabola passes through the points $(-\pi, 0)$ and $(\pi, 0)$ as shown.

If a is the focal length of the parabola:

(i) Show that $4a = \frac{\pi^2}{\alpha}$. 2

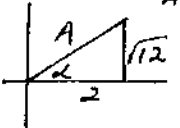
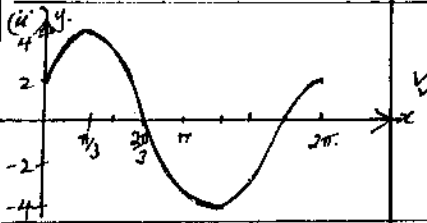
(ii) Show that $f(x)$ can be expressed in the form $f(x) = \alpha \left(1 - \frac{x^2}{\pi^2} \right)$. 2

(iii) Find the exact value of α given that the area between $y = f(x)$ and the x axis from $x = -\pi$ to $x = \pi$ is 4 square units. 3

- (b) Assume that tides rise and fall in Simple Harmonic Motion. A ship needs 11 metres of water to pass down a channel safely. At low tide, the channel is 8m deep and at high tide 12 m deep. Low tide is at 10:00 am and high tide at 4:00 pm. 5

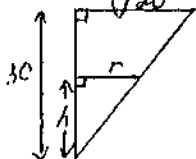
Find the first time period during which the ship can safely proceed through the channel.

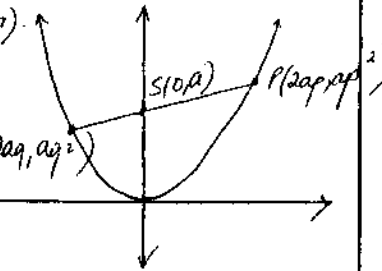
END OF PAPER

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
$1) {}^{12}C_3 (2)^9 \left(\frac{-2}{x^3}\right)^3$ $= {}^{12}C_3 \times \frac{(-2)^3}{(x^3)^3}$ $= {}^{12}C_3 \times (-2)^3$ $= -1760.$	✓	$\therefore \theta = 74^\circ 45'$ $\therefore \text{obscure } \angle = 105$	✓
$2) y = x^2 - x$ $\frac{dy}{dx} = 2x - 1$ $\text{@ } x = 2 \frac{dy}{dx} = 3.$ $\therefore y - y_1 = m(x - x_1)$ $y - 2 = 3(x - 2)$ $y = 3x - 6 + 2$ $y = 3x - 4$	✓	$c) (i) \sqrt{12} \sin x + 2 \cos x \equiv A \cos(x - \alpha)$ $\equiv A \cos x \cdot \cos \alpha + A \sin x \sin \alpha$ $\therefore A \cos \alpha = 2 \quad A \sin \alpha = \sqrt{12}$ $\cos \alpha = \frac{2}{A} \quad \sin \alpha = \frac{\sqrt{12}}{A}$  $A = 4.$ $\tan \alpha = \frac{\sqrt{12}}{2}$ $\tan \alpha = \sqrt{3}$ $\therefore \alpha = \frac{\pi}{3}$ $\sqrt{12} \sin x + 2 \cos x = 4 \cos\left(x - \frac{\pi}{3}\right)$	✓
$3) \frac{x}{3} + \frac{y}{2} = 1$ $2x + 3y = 6$ $3y = 6 - 2x$ $y = 2 - \frac{2}{3}x$ $\therefore m_1 = -\frac{2}{3}.$ $m_2 = 3.$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $\tan \theta = \left \frac{3 - (-\frac{2}{3})}{1 + 3(-\frac{2}{3})} \right $ $\tan \theta = 3^{\frac{2}{3}}$	✓		✓
		$(ii) 2$	✓
		$(iv) 4 \cos(x - \pi/3) = 1$ $\cos(x - \pi/3) = \frac{1}{4}.$ $\therefore x = \frac{\pi}{3} + 2\pi \pm \cos^{-1}\left(\frac{1}{4}\right)$	✓

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
$4) e^{-x} \log_e x = 0$ $b) f(x) = e^{-x} \log_e(x)$ $f'(x) = -e^{-x} - \frac{1}{x}$ $f(1.4) = e^{-1.4} - \log_e(1.4)$ $f'(1.4) = -e^{-1.4} - \frac{1}{1.4}$ $\text{hence } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $= 1.4 - \frac{e^{-1.4} - \ln(1.4)}{-e^{-1.4} - \frac{1}{1.4}}$ $\approx 1.306 \text{ (3dp)}$	✓	$= 42P - 35 + 5$ $= 42P - 30$ $= 6(7P - 5)$ $= 6Q; \text{ where } Q = 7P - 5$ $\text{which is divisible by 6.}$ If the statement is true for $n = k$, then the statement is true for $n = k + 1$. Since the statement is true for $n = 1$, then it is true for $n = 1 + 1 = 2$, $2 + 1 = 3$, etc for all positive integers n. note: students must have attempted steps 1, 2, 3 to be awarded marks for step 4.	✓
$b) \text{Test that the statement is true for } n = 1; \text{ where } n \text{ is a positive integer.}$ ie $7 + 5 = 12 = 6 \times 2$ divisible by 6. Assume that the statement is true for $n = k$, ie $7^k + 5 = 6P; \text{ where } P \text{ is a positive integer.}$ Prove that the statement is true for $n = k + 1$. ie $7^{k+1} + 5 = 6Q$; where Q is a positive integer. So $7^{k+1} + 5$ $= 7(7^k) + 5$ $= 7(6P - 5) + 5; \text{ from the assumption}$	✓	$c) \frac{d}{dx} \left(x \sin^{-1} \frac{x}{4} + \sqrt{16 - x^2} \right)$ $= x \cdot \frac{1}{\sqrt{16 - x^2}} + \sin^{-1} \left(\frac{x}{4} \right) \cdot 1 + \frac{1}{2} (16 - x^2)^{-\frac{1}{2}} \cdot (-2x)$ $= \frac{x}{\sqrt{16 - x^2}} + \sin^{-1} \left(\frac{x}{4} \right) + \frac{-x}{\sqrt{16 - x^2}}$ $= \sin^{-1} \left(\frac{x}{4} \right)$ $(ii) \int_0^4 \sin^{-1} \left(\frac{x}{4} \right) dx = \left[x \sin^{-1} \left(\frac{x}{4} \right) + \sqrt{16 - x^2} \right]_0^4$ from (i) $= (4 \sin^{-1}(1) + \sqrt{16 - 16}) - (0 + \sqrt{16})$ $= 4 \sin^{-1}(1) - 4$ $= 2\pi - 4.$	✓

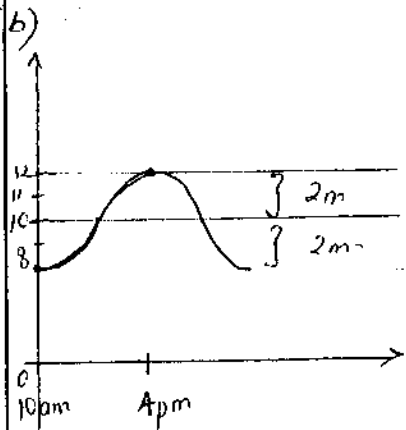
Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
when $t=0$, $T=24^{\circ}\text{C}$, $\theta = -40^{\circ}\text{C}$, $\therefore 24 = -40 + Ae^{\theta}$ $\therefore A = 64$	✓	$\text{LHS} = \frac{\tan \frac{\pi}{4} + \tan \theta}{1 - \tan \frac{\pi}{4} \cdot \tan \theta} - \left(\frac{\tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \cdot \tan \theta} \right)$ $= \frac{1 + \tan \theta}{1 - \tan \theta} - \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right)$ $= \frac{(1 + \tan \theta)^2 - (1 - \tan \theta)^2}{(1 - \tan \theta)(1 + \tan \theta)}$ $= \frac{1 + 2 \tan \theta + \tan^2 \theta - (1 - 2 \tan \theta + \tan^2 \theta)}{(1 - \tan \theta)(1 + \tan \theta)}$ $= \frac{1 + 2 \tan \theta + \tan^2 \theta - 1 + 2 \tan \theta - \tan^2 \theta}{(1 - \tan \theta)(1 + \tan \theta)}$ $= \frac{4 \tan \theta}{1 - \tan^2 \theta}$ $= \frac{2(2 \tan \theta)}{1 - \tan^2 \theta}$ $= \frac{2(\tan \theta + \tan \theta)}{1 - (\tan \theta)(\tan \theta)}$ $= 2(\tan 2\theta)$ $= \text{RHS.}$	✓
) $t = 5$, $T = 19^{\circ}\text{C}$. $19 = -40 + 64e^{5k}$ $e^{5k} = \frac{59}{64}$ $\ln(e^{5k}) = \ln\left(\frac{59}{64}\right)$ $\therefore 5k = \ln\left(\frac{59}{64}\right)$ $k = \frac{1}{5} \ln\left(\frac{59}{64}\right)$	✓		
) when $T = 0^{\circ}\text{C}$; $0 = -40 + 64e^{kt}$ $\frac{40}{64} = e^{kt}$ $\ln\left(\frac{40}{64}\right) = \ln(e^{kt})$ $\therefore kt = \ln\left(\frac{40}{64}\right)$ $t = \frac{\ln\left(\frac{40}{64}\right)}{k}$ $t = \frac{\ln\left(\frac{40}{64}\right)}{\frac{1}{5} \ln\left(\frac{59}{64}\right)}$ $t = 28.889 \dots$ $\approx 29 \text{ seconds}$	✓		✓

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
5c(i) using similar triangles:  $\therefore \frac{r}{h} = \frac{20}{30}$ $\frac{r}{h} = \frac{2}{3}$ $\therefore r = \frac{2}{3}h$	✓	$A = \pi r^2$ $A = \pi \left(\frac{2h}{3}\right)^2$ $= \frac{4\pi}{9} h^2$ $\frac{dA}{dh} = \frac{8}{9} \pi h$ $\therefore \frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt}$ $= \frac{8\pi h}{9} \times \frac{54}{\pi h^2}$ when $h = 16 \text{ cm}$ $\frac{dA}{dt} = \frac{8\pi}{9} \times \frac{54}{\pi 16}$ $= 3 \text{ cm}^2/\text{s}.$	✓
(ii) $V = \frac{1}{3} \pi r^2 h$; $r = \frac{2}{3}h$ $= \frac{1}{3} \pi \left(\frac{2h}{3}\right)^2 h$ $= \frac{1}{3} \pi \left(\frac{4h^2}{9}\right) h$ $= \frac{4}{27} \pi h^3$	✓		
(iii) $\frac{dV}{dh} = \frac{4}{9} \pi h^2$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{9}{4\pi h^2} \times 24$ $\frac{dh}{dt} = \frac{54}{\pi h^2}$			

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
 $pq = \frac{ap^2 - aq^2}{2ap - 2aq}$ $= \frac{a(p-q)(p+q)}{2a(p-q)}$ $= \frac{p+q}{2}$ $-ap^2 = \frac{p+q}{2}(x-2ap)$ $y-2ap^2 = (p+q)(x-2ap)$ $2y-2ap^2 = px - 2ap^2 + qx - 2apq$ $2y = (p+q)x - 2apq$ $y = \frac{1}{2}(p+q)x - apq$ Since PQ is a focal chord it passes through S(0, a). ✓ $a = \frac{1}{2}(p+q)(0) - apq$ $a = -apq$ $\therefore pq = -1.$	✓	iii) $PS = \sqrt{(2ap-0)^2 + (ap^2-a)^2}$ $= \sqrt{(2ap)^2 + a^2(p^2-1)^2}$ $= \sqrt{4a^2p^2 + a^2(p^4 - 2p^2 + 1)}$ $= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$ $= \sqrt{a^2(p^4 + 2a^2p^2 + 1)}$ $= \sqrt{a^2(p^2+1)^2}$ $= a(p^2+1)$ Similarly $QS = a(q^2+1)$ $PQ = PS + SQ$ $= a(p^2+1) + a(q^2+1)$ $= a(p^2 + q^2 + 2)$ Since $pq = -1$ $q = \frac{-1}{p}$ $\therefore PQ = a(p^2 + \frac{1}{p^2} + 2)$ $= a(p + \frac{1}{p})^2$	✓

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
6b(i) Using the chain Rule; $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dt} \left(\frac{1}{2} v^2 \right) \times \frac{dt}{dx}$ $= v \times \frac{dv}{dx}$ $= \frac{dx}{dt} \times \frac{dv}{dx}$ $= \frac{dv}{dt}$ $= \frac{d^2x}{dt^2}$	✓		
ii) $\frac{d^2x}{dt^2} = -4 \left(x + \frac{16}{x^3} \right)$ $\therefore -4 \left(x + \frac{16}{x^3} \right) = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $\frac{1}{2} v^2 = \int (-4x - 64x^{-3}) dx$ $\frac{1}{2} v^2 = \frac{-4x^2}{2} - \frac{64x^{-2}}{-2} + C$ $t=0, v=0, x=2, C=0.$ $\therefore v^2 = -4x^2 + 64x^{-2}$ $v^2 = \frac{64}{x^2} - 4x^2$ $v^2 = \frac{64 - 4x^4}{x^2}$ $v^2 = \frac{4(16 - x^4)}{x^2}$	✓		

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
$(x-h)^2 = -4a(y-k)$ vertex $(0, \alpha)$. $(x-0)^2 = -4a(y-\alpha)$ $x^2 = -4a(y-\alpha)$ function passes through $(\pi, 0)$ $\pi^2 = -4a(0-\alpha)$ $\pi^2 = -4a(-\alpha)$ $\pi^2 = 4a\alpha$ $\therefore 4a = \frac{\pi^2}{\alpha}$	✓	iii) $\int_{-\pi}^{\pi} \alpha \left(1 - \frac{x^2}{\pi^2}\right) dx = 4$ $2 \int_0^{\pi} \alpha \left(1 - \frac{x^2}{\pi^2}\right) dx = 4$ ✓ $\int_0^{\pi} \left(\alpha - \frac{\alpha}{\pi^2} x^2\right) dx = 2$ $\left[\alpha x - \frac{\alpha}{\pi^2} \frac{x^3}{3}\right]_0^{\pi} = 2$ $\left(\alpha\pi - \frac{\alpha}{\pi^2} \frac{\pi^3}{3}\right) - (0-0) = 2$ ✓ $\alpha\pi - \frac{\alpha\pi}{3} = 2$ $\frac{2}{3}\alpha\pi = 2$ $\alpha\pi = 3$ $\alpha = \frac{3}{\pi}$ ✓	
$x^2 = -4a(y-\alpha)$ since $4a = \frac{\pi^2}{\alpha}$ $\therefore x^2 = -\frac{\pi^2}{\alpha}(y-\alpha)$ $x^2 = -\frac{\pi^2}{\alpha}y + \pi^2$ $\alpha x^2 = -\pi^2 y + \alpha\pi^2$ $\pi^2 y = \alpha\pi^2 - \alpha x^2$ $y = \frac{\alpha\pi^2 - \alpha x^2}{\pi^2}$ $y = \alpha \left(1 - \frac{x^2}{\pi^2}\right)$	✓		

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
7b)  10am 4pm 6 hours. $\therefore \text{wavelength} = 12\text{h.}$ $\text{period} = \frac{2\pi}{n} = 12$ $\therefore n = \frac{\pi}{6}$ Amplitude = 2m. $x = -2 \cos\left(\frac{\pi}{6}t\right) + 10$ when $x = 11\text{m};$ $11 = -2 \cos\left(\frac{\pi}{6}t\right) + 10$ $-\frac{1}{2} = \cos\left(\frac{\pi}{6}t\right)$ $\therefore \frac{\pi}{6}t = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$ $t = \frac{2\pi \times 6}{3 \times \pi}, \frac{4\pi \times 6}{3 \times \pi}$ $= 2\text{ hours.}$ $t = 4\text{ hours, } 8\text{ hours}$ from 10am.	✓ ✓ ✓ ✓ ✓	ie, the first time period the ship can safely pass through would be between 2pm and 6pm.	